Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

" To elimination the concern of Brake oil filling machine failure at Scorpio TCF Final line"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**06.04.19**	**8.11.18**
Completion Date	**23.04.19**	**27.11.18**

Current Status: **CLOSED**

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Dattatray Jondhale	L-7 O	STCF	7709341353
2	**Mentor**	Tushar Patil	L-5 M	STCF	9922409136
3	**Member 1**	Devre Nilesh	L-8 O	STCF	8888821705
4	**Member 2**	Khade Hemant	L-8 O	STCF	9766874268
5	**Member 3**	Kekade Sagar	L-8 O	STCF	8237028924
6	**Member 4**	Sirvastava Prateek	L-8 O	STCF	7007079219
6	**Member 5**	Satish Pagar	L-8 O	STCF	9822209653

Step1. Select Subject

PROJECT TITLE :-

To elimination the concern of Brake oil filling machine failure at Scorpio TCF Final line

Brief Description :

- **What** : FRC Loss at brake oil filling machine
- **Where** : At Final line , Filling station
- **When** : Since Aug-18
- **Who** : by Final Line Associate & officers
- **Why** : Major irritant on line, FRC loss.
- **How** : **9** avg. incidences per month FRC loss due to brake oil filling machine not working properly

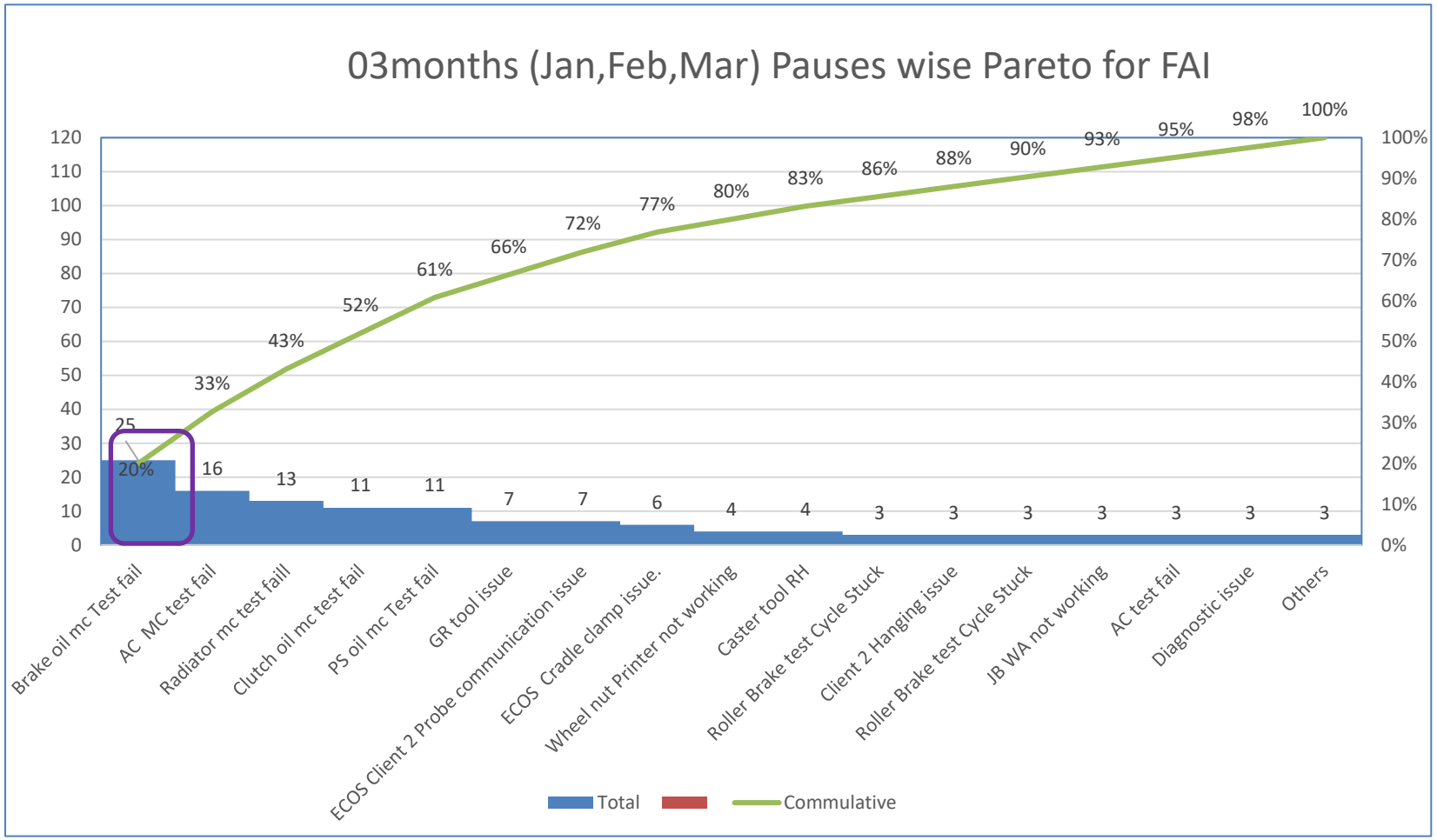
Brake oil filling machine

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Incidences →

Concern →

<u>INFERENCE :-</u>AS PER PARETO GRAPH Brake oil filling machine ISSUE IS TOP in minor stoppages list

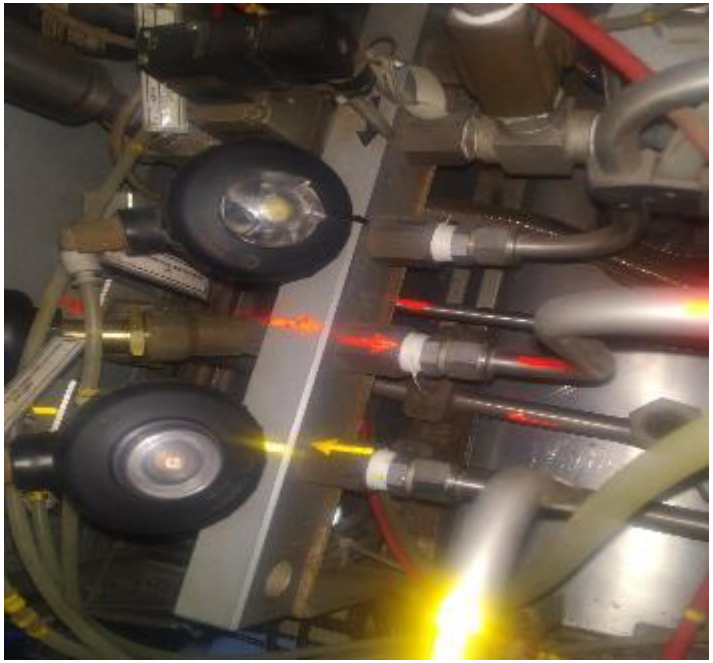
2. Justify the choice

COPQ Cost Calculation -

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	0.5
Total Required Rework time per day	Min	15
Effective working time	Min	465
Operator required /day	nos	0.03
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	63.68
Avg no. of working days	days	25
Operator cost /month	Rs	1591.9
Operator Cost/year	**Rs**	**19103.2**

2.b Immediate Containment Action

- oil filling Gun replacement
- valve replacement



Gun repairing and replacement

Valve replacement

Step 3. Understanding Current Situation

Observation of process:-

Vehicle At Brake oil filling machine

Stage associate scan vehicle

Cycle selection done on machine

Associate connect OBD Cable to vehicle & lift gun from holder and clamp it to vehicle

DEFECT GENERATION STAGE

Associate start cycle start button on gun

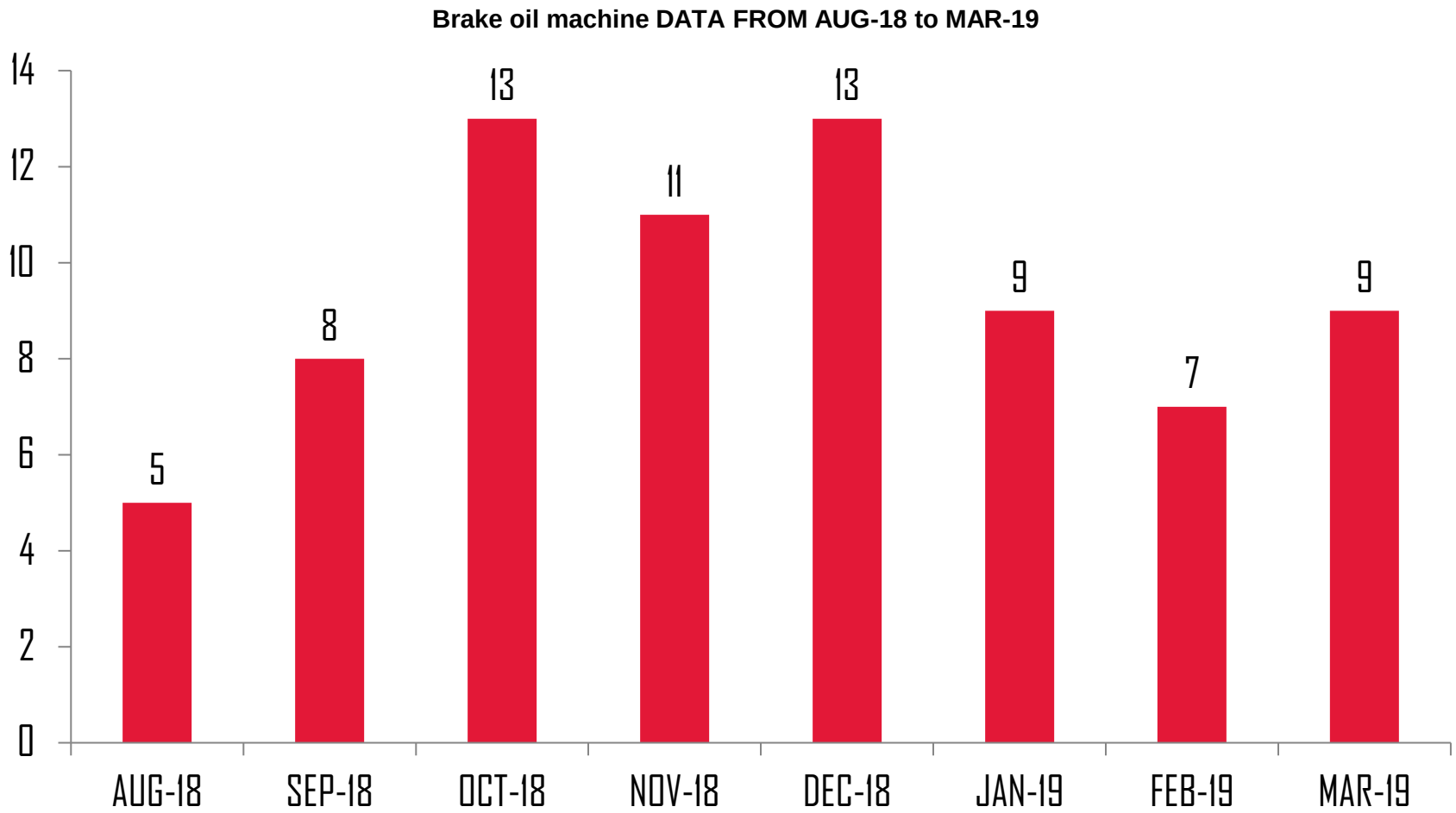
Filling process start After filling Associate press stop button and gun put in holder

DEFECT GENERATION STAGE

OK vehicle to next stage

3. Understand the current situation

Month wise Trend

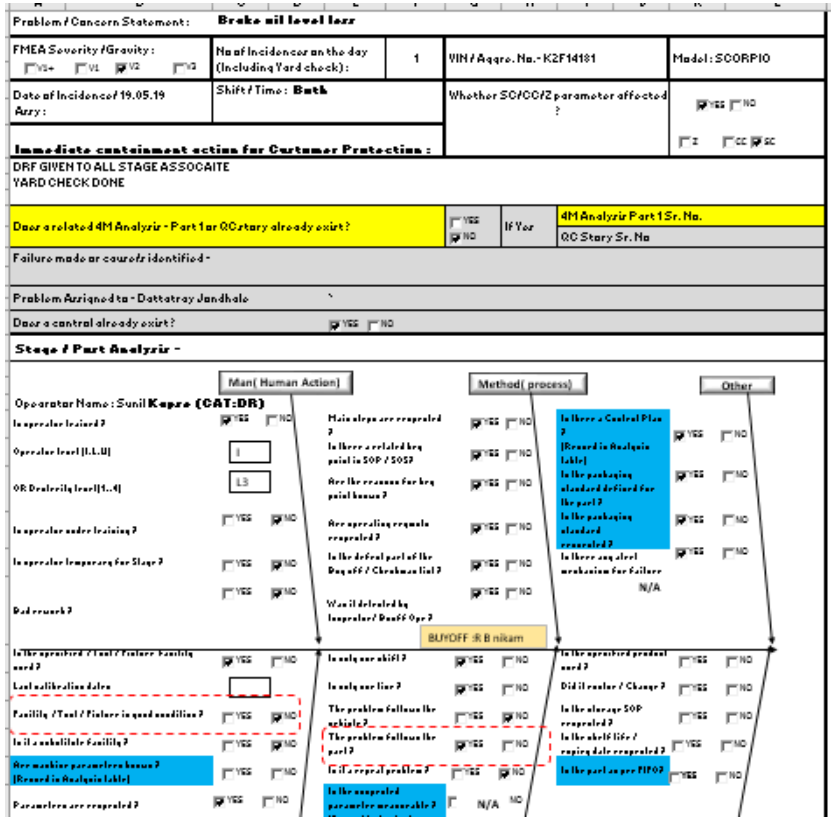


INFERENCE :- AS PER MONTH ON MONTH DATA COLLECTED THE SAID DEFECT IS CONTRIBUTING AVERAGE 9 concerns/month

Step 3. Understanding Current Situation

Good & Not So good observations

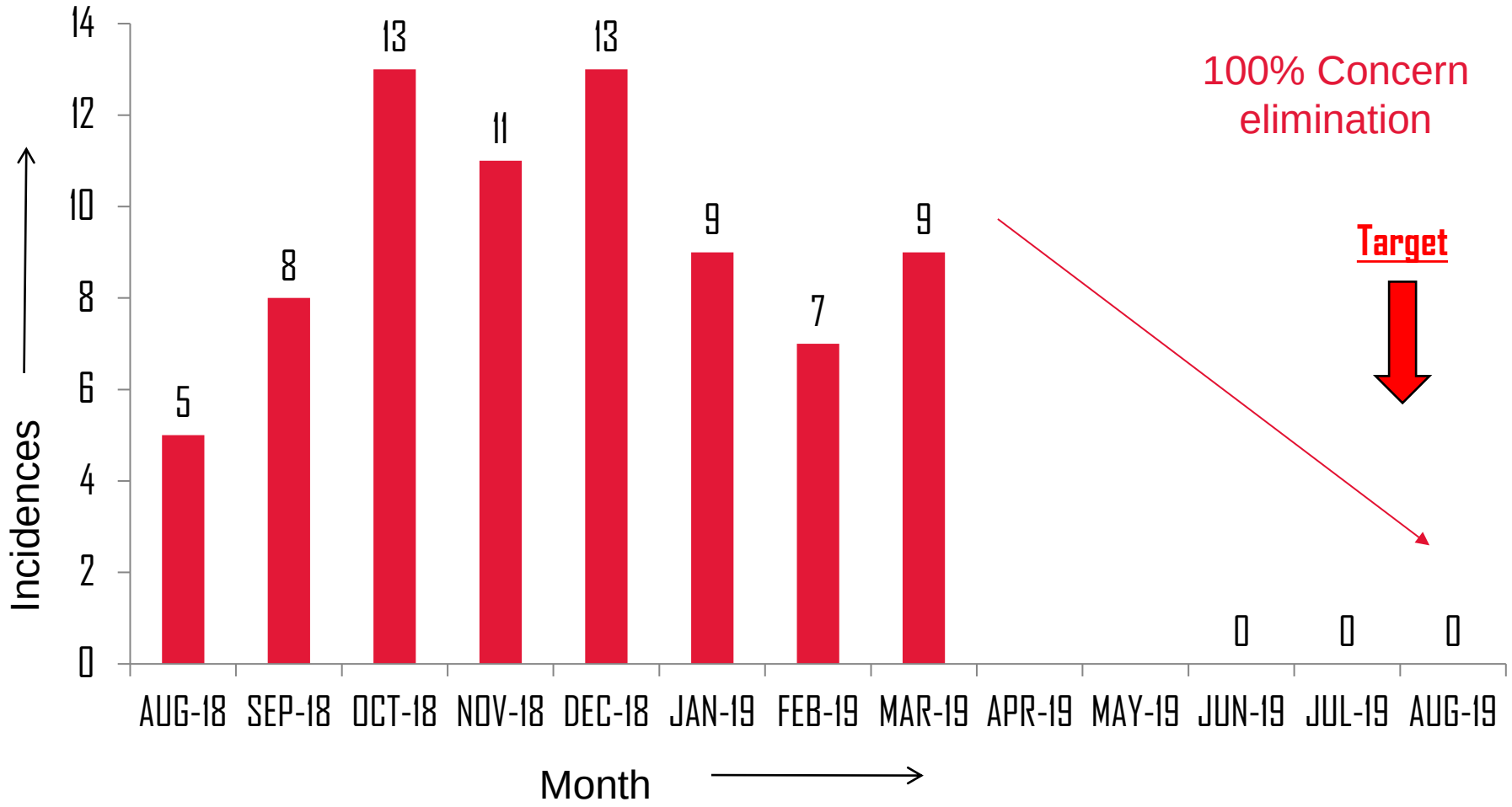
- Issue Resolve within 10 to 15 min.
- Issue reported in both the shifts
- Issue occurs frequently
- Reasons for stoppage is similar



4. Select targets

Present level :- 105 cases reported in last 10 months

Target :- Concern elimination by END OF AUG-2019



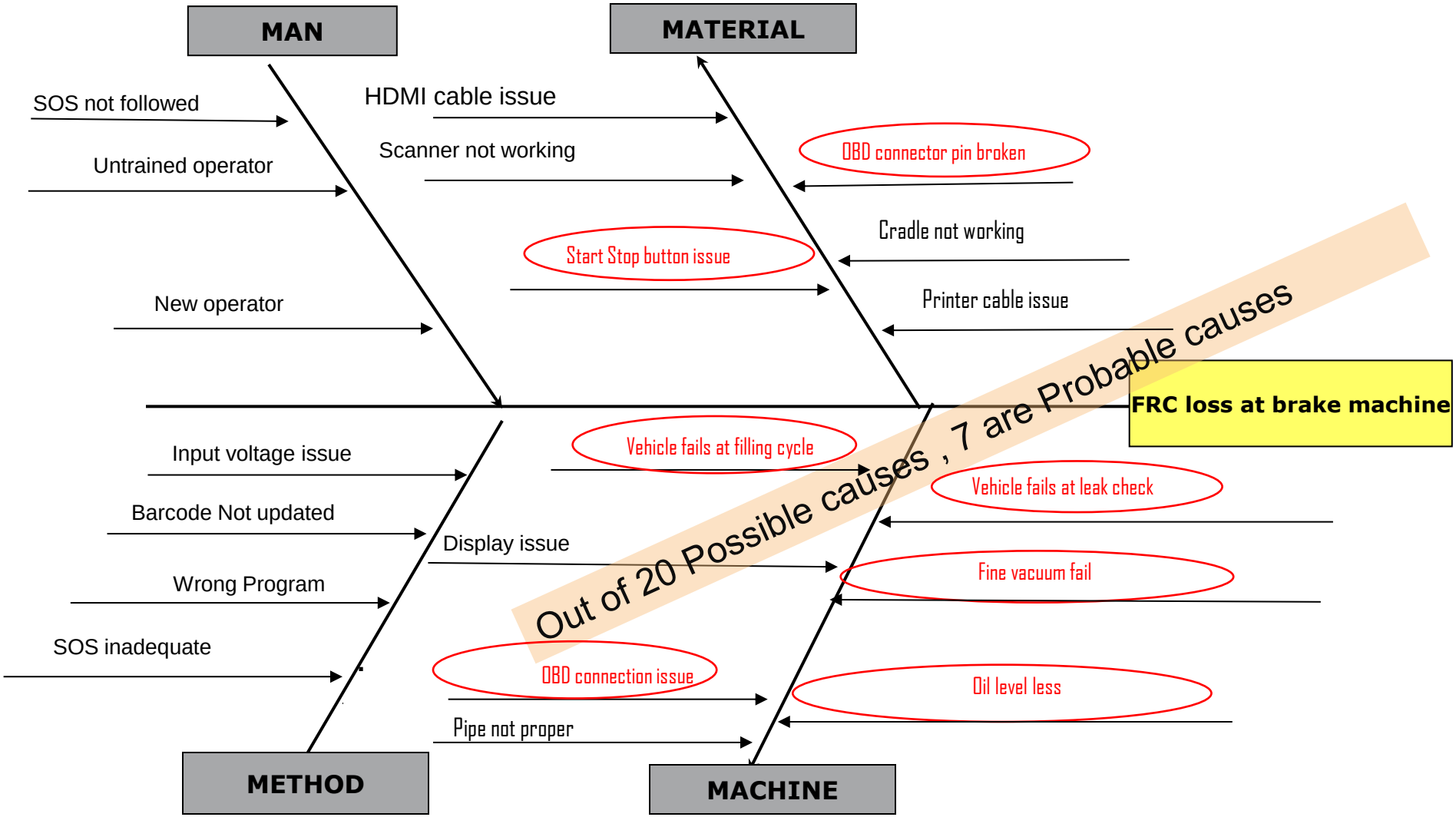
Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible cause	Sr. no	Possible cause
1	New Operator	11	OBD pin Broken
2	Untrained operator	12	Vehicle fails at leak check
3	SOS not followed	13	Vehicle fails at filling test
4	SOS inadequate	14	Display Issue
5	Pipe not proper	15	Start/ stop button issue
6	Scanner not working	16	Oil level less
7	Cradle not working	17	Program Wrong
8	Fine Vacuum Test fail	18	Barcode not updated
9	Input voltage issue	19	Printer cable issue
10	OBD connector issue	20	HDMI cable issue

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Dattatray	Tushar	Nilesh	Hemant	Sagar	Prateek	Satish	Total
1	New Operator	1	0	0	0	3	3	3	10
2	Untrained operator	3	3	0	0	3	3	3	15
3	SOS not followed	3	3	0	0	3	1	0	10
4	SOS inadequate	9	0	0	1	3	0	0	13
5	Pipe not proper	3	3	0	0	3	3	3	15
6	Scanner not working	3	0	0	0	3	0	3	9
7	Cradle not working	0	0	3	0	3	3	0	9
8	Fine Vacuum Test fail	9	9	3	9	3	9	3	45
9	Input voltage issue	3	3	3	3	1	0	0	13
10	OBD connector issue	3	9	3	9	9	3	9	45
11	OBD connector PIN broken	9	3	9	9	3	3	3	39
12	Vehicle fails at leak check	9	3	3	9	3	9	9	45
13	Vehicle fails at filling test	9	3	9	9	3	3	3	39
14	Display issue	3	1	3	0	0	0	3	10
15	Oil level less	9	9	3	3	9	9	3	45
16	Start / Stop button issue	9	9	9	9	3	3	9	51
17	Program Wrong	3	0	3	1	3	3	3	16
18	Barcode not updated	3	1	3	1	3	1	3	15
19	Printer cable issue	1	0	3	3	0	3	1	11
20	HDMI cable issue	0	0	3	3	0	0	0	6

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Not considered Causes	Reasons
1	New Operator	NO New operator deployed on this stage
2	Untrained operator	All trained operator
3	SOS not followed	SOS followed
4	SOS inadequate	SOS was OK
5	Pipe Not inserted	In audit found ok
6	Scanner not working	Scanner found ok no issue
7	Cradle not working	Cradle found ok no issue
8	Input voltage issue	Voltage as per Specification
9	Display issue	Found ok
10	Program Wrong	Program OK
11	Barcode Not updated	All Barcode updated
12	Printer Cable issue	Cable continuity ok
13	HDMI Cable issue	Cable continuity ok

5.1 Identify Probable cause

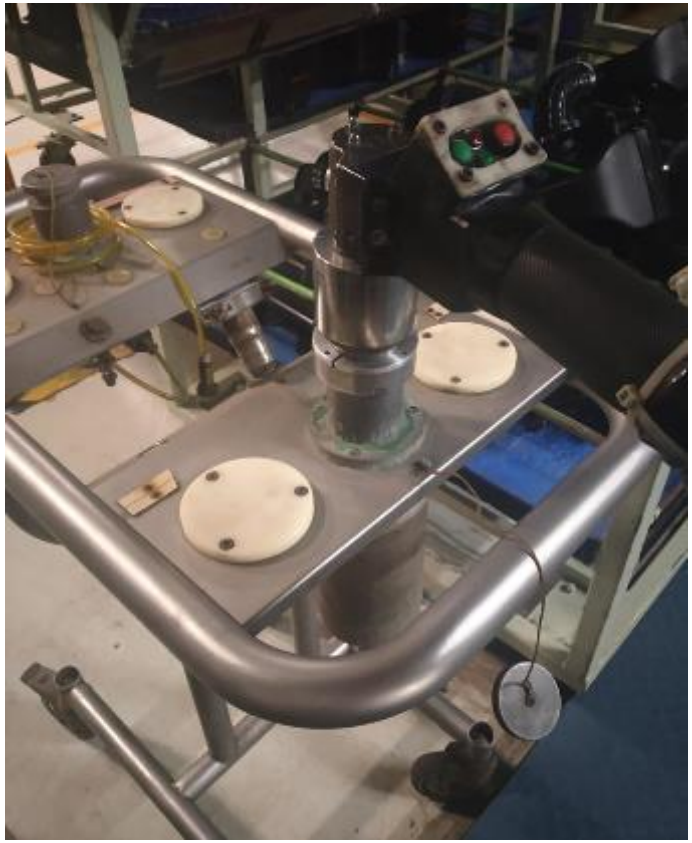
No	List of Probable Causes
1	Fine Vacuum test fail
2	OBD connector issue
3	OBD Pin broken
4	Vehicle fails at leak check
5	Vehicle fails at filling test
6	Oil level less
7	Start / Stop button issue

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Fine vacuum test fail	Checking of Specification and confirm whether it is as per Requirement or not. Limit is 2.2mb	Trial taken on test pot found ok	**Hypothesis invalid**

Test pot trial for 14 cycle			
Sr.No.	Coarse Vacuum	Fine Vacuum	Fine vacuum limit
1	8.7	1.8	2.2
2	9.5	1.9-2.1	2.2
3	9.3	1.9-2.0	2.2
4	7.1	1.9	2.2
5	8.8	1.8	2.2
6	7.5	1.8	2.2
7	9.6	1.9-2.0	2.2
8	8	1.8	2.2
9	8.8	1.8	2.2
10	6.4	1.9	2.2
11	7.3	1.9	2.2
12	6.6	1.9	2.2
13	6.4	1.8	2.2
14	9.1	1.9-2.0	2.2

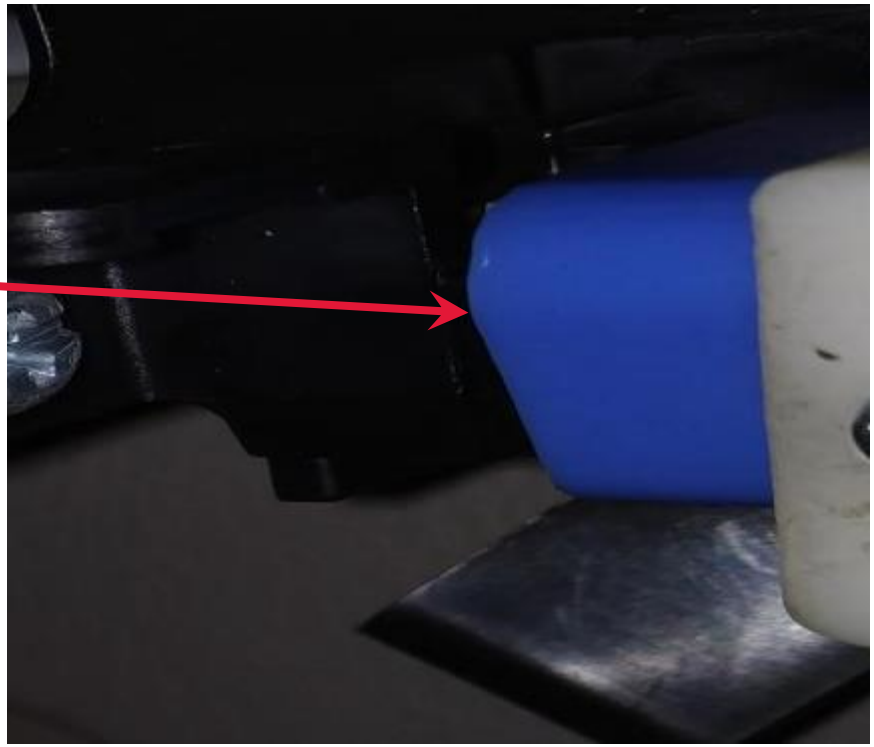


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	OBD connection Issue	Visual Check of OBD connector and confirm whether it Properly inserted or not.	Connector checked found – Properly inserted Specifications = it should inserted properly	**Hypothesis invalid**

Connector connected properly

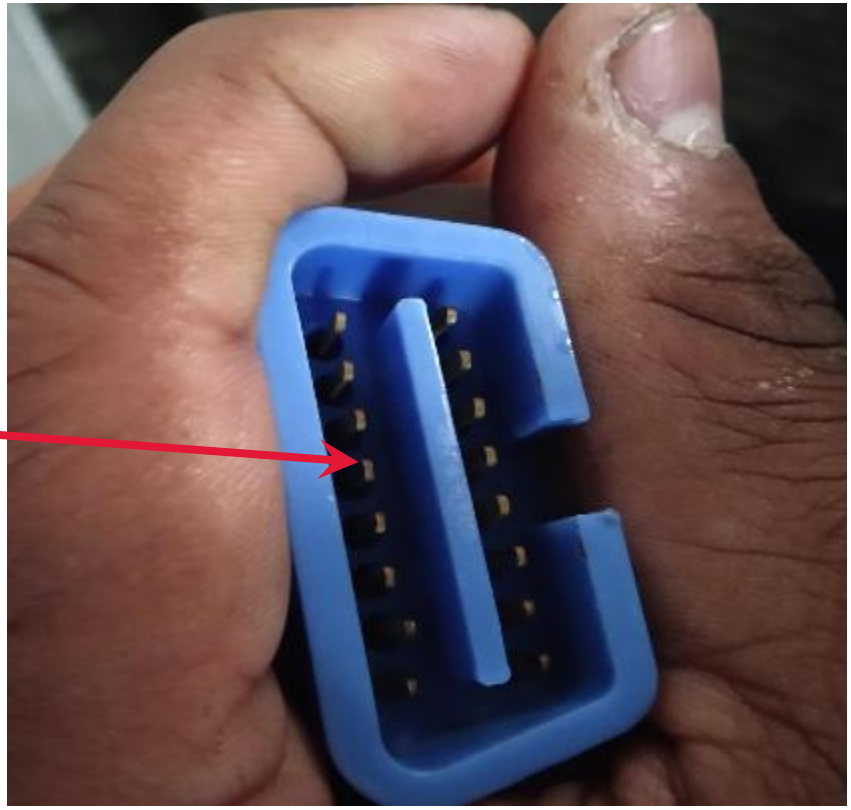


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	OBD pin broken	Checking visually for All 16 pin present or not	Connector checked found – All 16 pin properly present Specifications – 16 pin should be there	**Hypothesis invalid**

All pins found ok



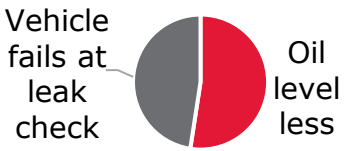
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Vehicle fails at leak check	Checking of Specification and confirm whether it is as per Requirement or not. Limit is 0.2mb	Observation noted in 10 trials on test pot 2 cycle fails at leak check	**Hypothesis valid**

Test pot trial for 10cycle					
Sr.No.	Coarse Vacuum	Fine Vacuum	Fine vacuum limit	Leak check	leak check limit
1	8.7	1.8	2.2	0	0.2
2	9.5	1.9-2.1	2.2	0.1	0.2
3	9.3	1.9-2.0	2.2	0	0.2
4	7.1	1.9	2.2	0.3	0.2
5	8.8	1.8	2.2	0.1	0.2
6	7.5	1.8	2.2	0.1	0.2
7	9.6	1.9-2.0	2.2	0.1	0.2
8	8	1.8	2.2	0.3	0.2
9	8.8	1.8	2.2	0.1	0.2
10	6.4	1.9	2.2	0.1	0.2

Last 1 Year Data analyze and Categories in different causes



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Vehicle fails at filling test	Checking for pressure Specification and confirm whether it is as per Requirement or not its limit is minimum 4.2 bar.	Observation noted in 10 trials on test pot found ok	**Hypothesis invalid**

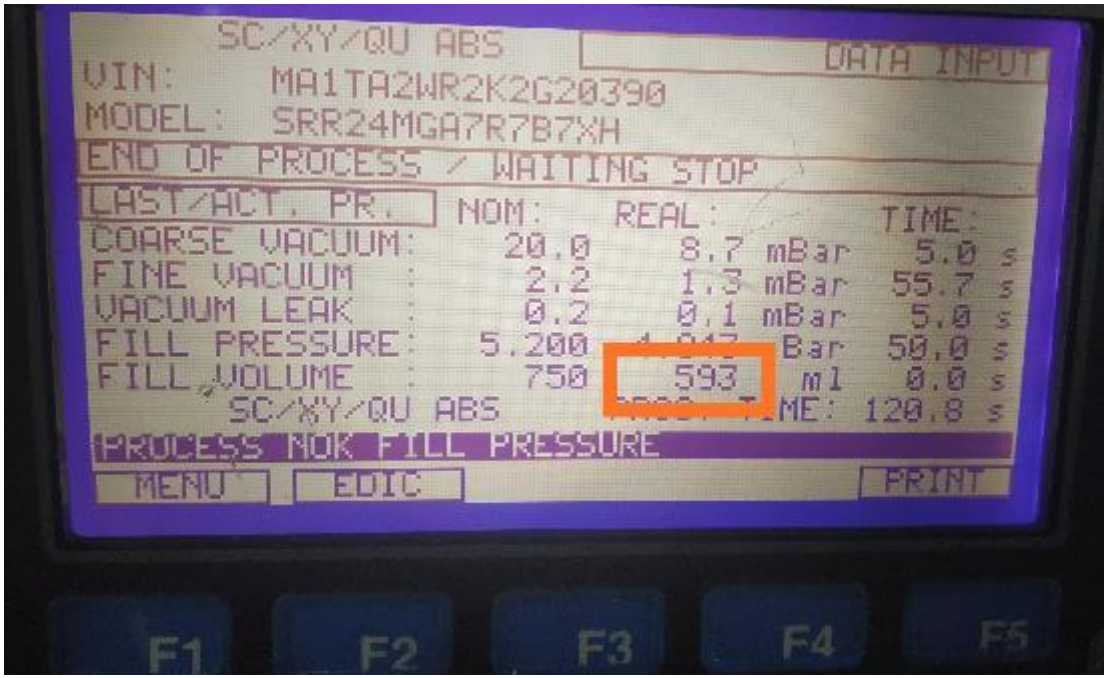
Date & Time	VIN NUMBER	MODEL NUMBER	COARSE VACUUM (mbar)	FINE VACUUM (mbar)	VACUUM LEAK (mbar)	FILL PRESSURE (bar)	FILL VOLUME (ml)	CYCLE TIME (sec)	RESULT
6/9/2019 11:26:38	MA1TA2WR2K2F12095	SRR24MFA7R7B6WP	6.5	1.7	0	4565	773	11515.00	PROCESS OK
6/9/2019 11:30:09	MA1TA2WR2K2F12096	SRR24MFA7R7B7WP	7.3	1.5	0	4575	773	11345.00	PROCESS OK
6/9/2019 11:34:21	MA1TA2SJXK2F12097	SFT24MCA9R7B7WD	6.5	1.5	0	4562	767	11009.00	PROCESS OK
6/9/2019 12:08:10	MA1TA2TDKK2F12098	SHG24MBA7R5B7WD	5.2	1.5	0	4211	683	10494.00	PROCESS OK
6/9/2019 12:12:08	MA1TA2SJXK2F12099	SFT24MCA7R7B7XH	6.3	1.4	0.1	4542	767	11053.00	PROCESS OK
6/9/2019 12:16:24	MA1TA2SJXK2F12101	SFT24MCA9R7B7WD	6.6	1.4	0	4534	768	10988.00	PROCESS OK
6/9/2019 12:20:15	MA1TA2SJXK2F12100	SFT24MCA7R7B7WD	7.2	1.4	0	4554	766	11134.00	PROCESS OK
6/9/2019 12:24:00	MA1TA2SJXK2F12102	SFT24MCA9R7B7WD	6.3	1.4	0	4533	596	11033.00	PROCESS OK
6/9/2019 12:27:59	MA1TA2SJXK2F12103	SFT24MCA9R7B7WD	6.9	1.4	0	4549	767	11084.00	PROCESS OK
6/9/2019 12:31:59	MA1TA2SJXK2F12104	SFT24MCA7R7B7XH	6.1	1.5	0	4531	601	11027.00	PROCESS OK
6/9/2019 12:35:46	MA1TA2SJXK2F12105	SFT24MCA7R7B7XH	7	1.5	0	4527	765	11213.00	PROCESS OK
6/9/2019 12:39:27	MA1TA2SJXK2F12106	SFT24MCA7R7B7XH	6.5	1.4	0	4525	598	11010.00	PROCESS OK

Pressure is more than 4.2 bar

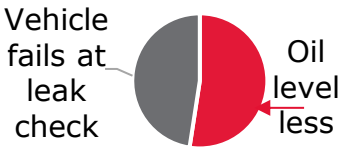
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	Oil level less	Checking Specification for Scorpio its limit is 750ml. and confirm whether it is as per Requirement or not	In analysis it is found that oil level varies from 584ml. To 664ml.	**Hypothesis valid**



Last 1 Year Data analyze and Categories in different causes



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Start / Stop button issue	Visual checking with OK ,NOT OK	checking done for button operation found ok	**Hypothesis Invalid**

checking done for button operation found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Vehicle fails at leak check	Checking of Specification and confirm whether it is as per Requirement or not. Limit is 0.2mb	**Hypothesis valid**
2	Oil level less	Checking Specification for Scorpio its limit is 750ml. and confirm whether it is as per Requirement or not	**Hypothesis valid**

2 Valid Cause

Analysis

5.2 Confirm the probable cause:

COMPARISON STUDY for Brake Machine working and non working condition

Brake machine not working	Brake machine working
CAD LEVEL COMPARISON	
➢ Pressure is more than 4.2 bar ➢ Vacuum also within a limit ➢ Oil level found 493ml.	➢ Pressure is more than 4.2 bar ➢ Vacuum also within a limit ➢ Oil level found 750ml.

CONCLUSION :- **Every component functioning but still machine shows low oil level**

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Vehicle fails at leak check	Gun not working properly	Seal damage	Time base maintenance frequency issue	TBM should be done on cycle base considering production volume
2	Oil level less	Oil filling valve close at filling cycle	SMC valve bank not sending signal properly	Valve bank position not fix properly so vibration when console move along with conveyor	1) Modify the current facility 2) Valve TBM done on cycle basis

6. Implement the corrective measures

Possible Remedies Identified

Three possible solutions identified to eliminate FRC loss at brake machine.

Legends
High ↑ Low ↓

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	**1) Vehicle fails at leak check** **2) Oil level less**	Put stand by MC OR replace all valve & Gun	High ↑	High ↑	No	High ↑	**Rejected**
		Modify the valve layout	Low ↓	High ↓	No	Low ↓	**Accepted**
		Modify the current TBM system with Digitization.	Low ↓	High ↓	No	Low ↓	**Accepted**

6. Implement the corrective measures

Possible Remedies Identified

SOLUTION PROPOSAL 1

1. Two Put stand by MC OR replace all valve & Gun

Two Machine to be put on stage or all valve and gun replacement leads to high cost

CONCLUSION :- This solution involve high cost and then also total problem will not be eliminated.

PROPOSAL REJECTED

6. Implement the corrective measures

Possible Remedies Identified

SOLUTION PROPOSAL 2

2. Modify the valve layout & TBM on cycle basis.

SMC valve bank position is not accessible so to check this valve more than 50min required as vacuum pump need to remove so this valve bank position shifted to accessible location so that its checking and replacement possible within 5min. Also valve and gun seal replacement should be done on cycle basis considering production volume for that one SCADA system to be develop which should give trigger of replacement time.

CONCLUSION :- This solution feasibility need to check from implementation point of view. This will eliminate Both the problems.

PROPOSAL ACCEPTED

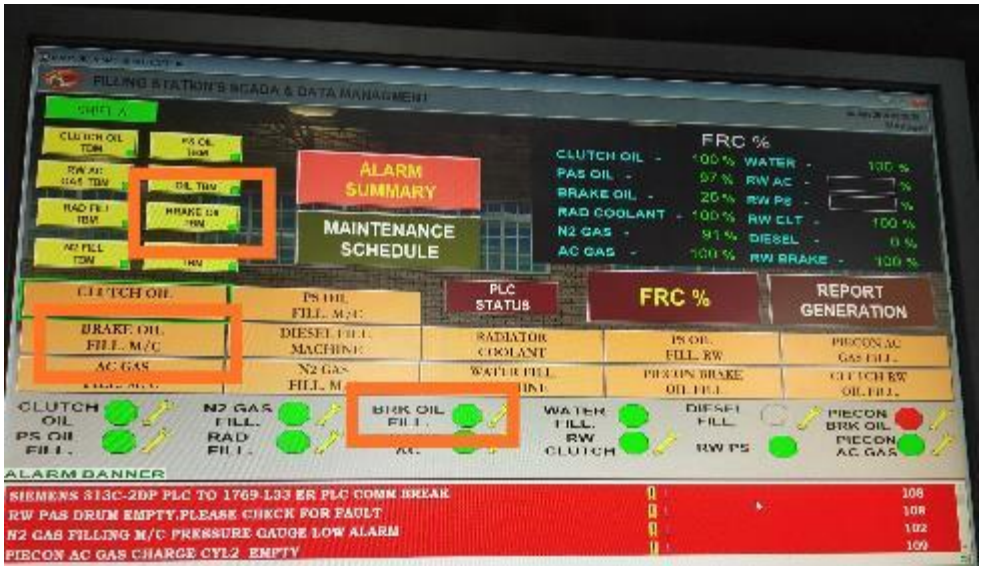
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Modify the current TBM system with Digitization. & Valve layout	Modify the Valve layout – SCADA installation done which gives online cycle base trigger to replace seals	04.06.2019	Dattatray Jondhale	CLOSED

Before

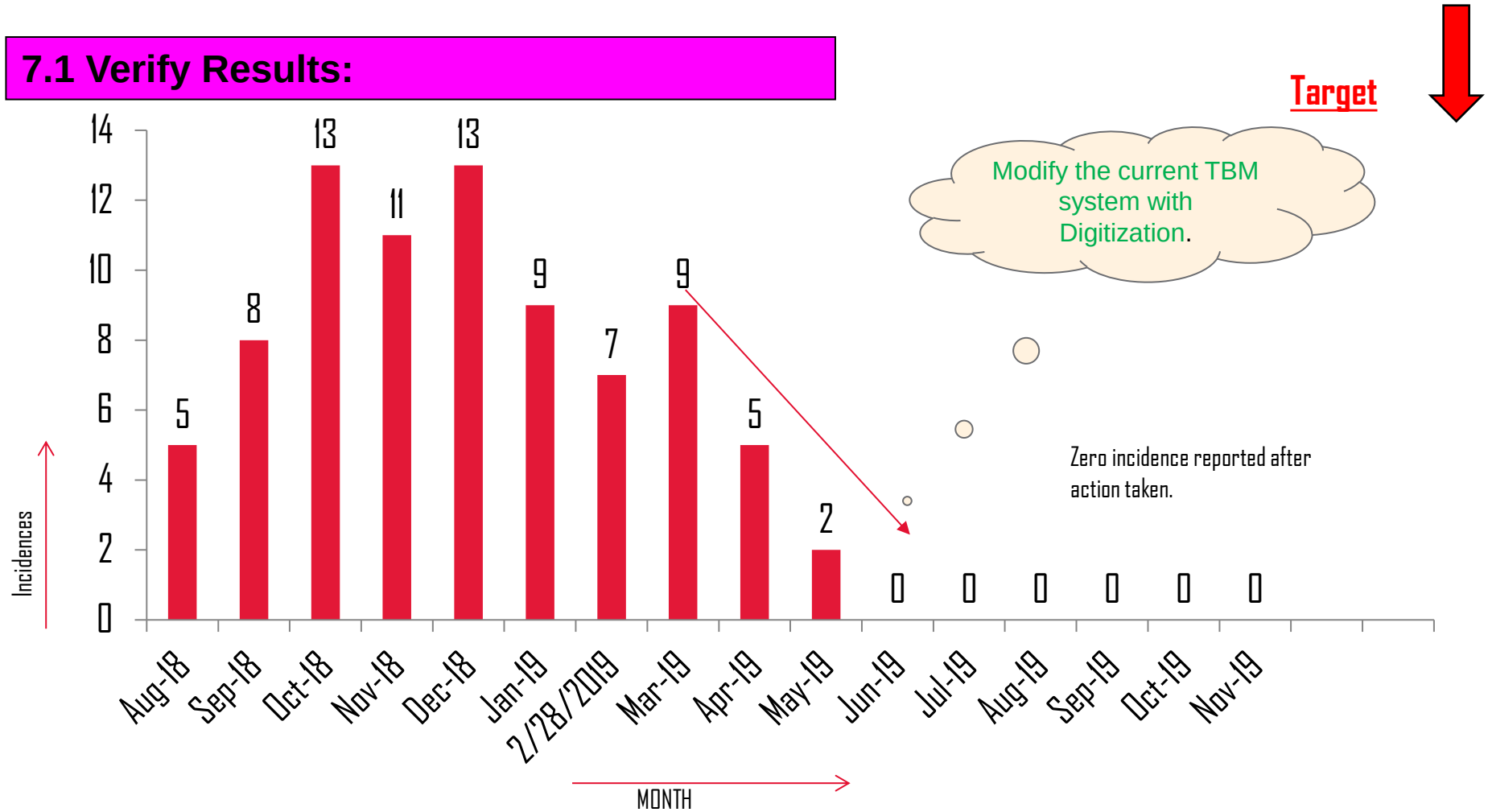
After – Valve position change



After – SCADA for online Cycle base TBM

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE :- **As per data collected ,after action implementation no defect reported in last 6 months. Hence concern is eliminated after taking PCA action through MYB project.**

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Design of New system	16.04.2019	Dattatray Jondhale	CLOSED
2	Development of system	06.05.2019	Dattatray Jondhale	CLOSED
3	Installation of SCADA	21.05.2019	Dattatray Jondhale	CLOSED
4	MP Sheet	09.07.2019	Hari Nagar	CLOSED
5	PM Sheets	8.07.2019	Dattatray Jondhale	CLOSED
6	Daily Check sheet	28.07.2019	Dattatray Jondhale	CLOSED

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) 100 FRC achieved for brake machine
2) Minor Stoppages of Brake machine eliminated.
2) Production loss due to Brake machine failure is eliminated completely.

In tangible Benefits –

1) Associate moral improved
2) Get direct benefits in RFD
3) Learn the Gemba culture

DR

Wed 1/8/2020 1:49 PM

DHANOKAR RAJESH

RE: MYB Dattatray Jondhale

To JONDHALE DATTATRAY--NASHIK

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	0.5
Total Required Rework time per day	Min	15
Effective working time	Min	465
Operator required /day	nos	0.03
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	63.68
Avg no. of working days	days	25
Operator cost /month	Rs	1591.9
Operator Cost/year	**Rs**	**19103.2**

From: JONDHALE DATTATRAY--NASHIK
Sent: Wednesday, January 08, 2020 7:32 AM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: MYB Dattatray Jondhale

Dear Sir,
Sending here with Cost Saving data for Rework of Brake oil test fail vehicle please help

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, Daily check sheet.

SN	Document	Actual date	Responsibility	Status
1	MP Sheets	09.07.2019	Hari Nagar	CLOSED
2	PM Sheets	8.07.2019	Dattatray Jondhale	CLOSED
3	Daily check sheet	28.07.2019	Dattatray Jondhale	CLOSED
4	CLITA Sheet updated	28.08.2019	Dattatray Jondhale	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Training given to Maintenance Associate.	07.08.2019	Dattatray Jondhale	Closed

8.3 Additional Audit & check

1	Check point added in DWM	01.08.19	Dattatray Jondhale	Closed

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

SN	Document	Actual date	Responsibility	Status
1	MP Sheet	09.07.2019	Hari Nagar	CLOSED

Gemba Kaizen Sheet

Plant :Nashik	PU / Shop / Dept. : SXTCF	Cell / Circle :
Machine / Equipment / Workstation No. :	Machine/Equipment/Workstation Name: Brake	Date : June2019

Pillar Applicability (pl. tick) :

KK / TM	JH	PM	QM	DM	E&T	OTPM	SHE

Theme : Ergonomics Kaizen to eliminate the bending operation

BEFORE	AFTER
Description in details :-	Description in details :-
SMC valve bank position is not accessible	valve bank position shifted to accessible location

KEY BENEFITS

Sr.no.	Description	Benefits
1	SMC valve bank position is not accessible so to check this valve more than 50min required as vacuum pump need to remove so this valve bank position shifted to accessible	BD time reduce by 45 min

Mahindra Rise. Automotive Sector Nasik-422007	Maintenance Prevention (MP) Sheet	PU/Dept : SXTCF Cell : Final Line
PU	Scorpio TCF	
Cell	BWT 12	
Machine	Brake oil filling machine	
Machine No.	PM0101929	
Phenomenon Observed	SCADA installation	
Date	09.07.19	
Why-Why Analysis	Vehicle test fail. ↓ Seal of gun & Valve issue ↓ Valve seal inproper TBM	
Root Cause	Production volume impact	
Immediate action taken to start machine/equipment	Installation of SCADA system	
Changes done in Design to eliminate breakdown	Cycle base TBM with online trigger to eliminate problem	

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

SN	Document	Actual date	Responsibility	Status
2	PM Check Sheet Updated	8.07.2019	Dattatray Jondhale	CLOSED

Preventive Maintenance Sheet — Frequency : 15 days

Area :- Final Line | Name of Machine :- Durr Brake Filling Machine (OFF LINE) | PM Date :

Sr.No	Item (What)	Activity	Method (How)	Required Value	C/NC	Remark
				Mechanical Panel		
1	PR oil	Inspect	Physical	Inspect for No oil / dust particles. Check that there is no condensation.	○	
2	Air line for all the valves	Inspect	Visual	Check all the air lines used as pilot air to operate the solenoid valve, Adapter open operating valve for leakage, bend, damge. If found repair the same.	○	
3	SBB Vacuum Pump	Inspect/ clean /Tighten	Physical /Allen key	Check the Vacuum pump oil level should be in between Min & MAX. Check the foundation of vacuum pump for lighening as per marking. Clean the vacuum pump & exhaust Fan. (MF * K.F – 24 Nm) (Durr Vacuum pump Grease oil DOT4 only)	○	
4	Vacuum Sensor & Pressure Sensor	Inspect /clean		Clean the sensor from outside.	○	
5	Exhaust filter	Clean	Physical	Filters should be clean	○	
6	Filling Pump	Inspect/ clean /Tighten	Physical /Allen key	Check the filling pump foundation for lighening as per marking. Clean exhaust Fan. Smooth operating. No leakage. (MF * K.F – 24 Nm)	○	
7	Check all solenoid valves	Inspect	Visual	Functioning ok Valves	○	
8	Check Non return valves	Inspect	Visual	No Oil leakage.	○	
9	Brake fluid reservior tank	Inspect /Tighten	Visual/ Allen key	Inspect the Reservior tank foundation bolt tightening as per marking. Check tank for no oil leakage. (MF*K.F – 24 Nm)	○	
10	Air Booster	Inspect	Visual	Inspect the Booster Output Air Pressure Dial Gauge (6 bar)	○	
11	Adapter clamp	Inspect/ clean	Phy. Check	Inspect adapter clamp & clean the clamp. Check for wear out. Check the condition of the felt seal replace as per TBM	○	
12	Adapter holder	Tighten	Physical	With spanner tighten all stud on the adapter, confirm No loose fitting.	○	
13	Adpter Hose pipe & Sleeve	Inspect	Visual	Check the hose pipe sleeve for damage.	○	

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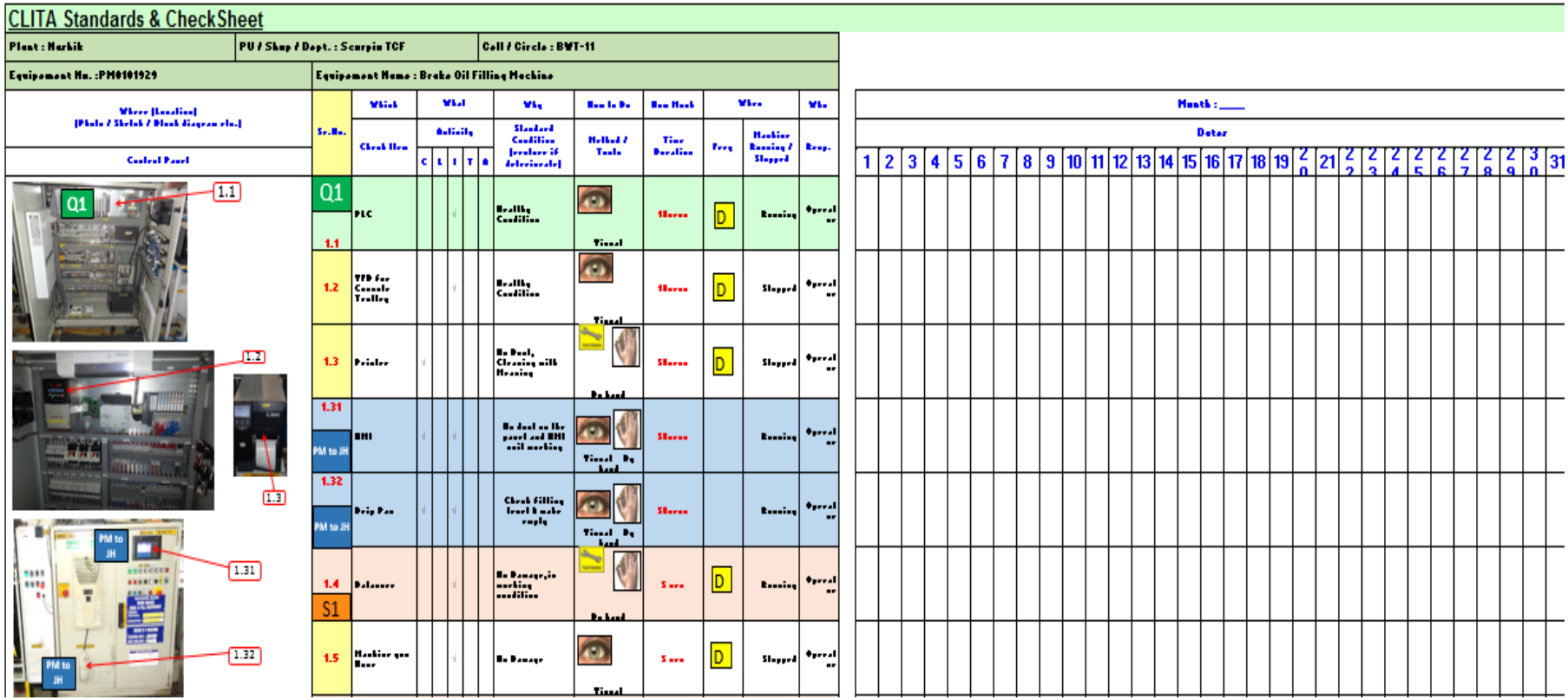
8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

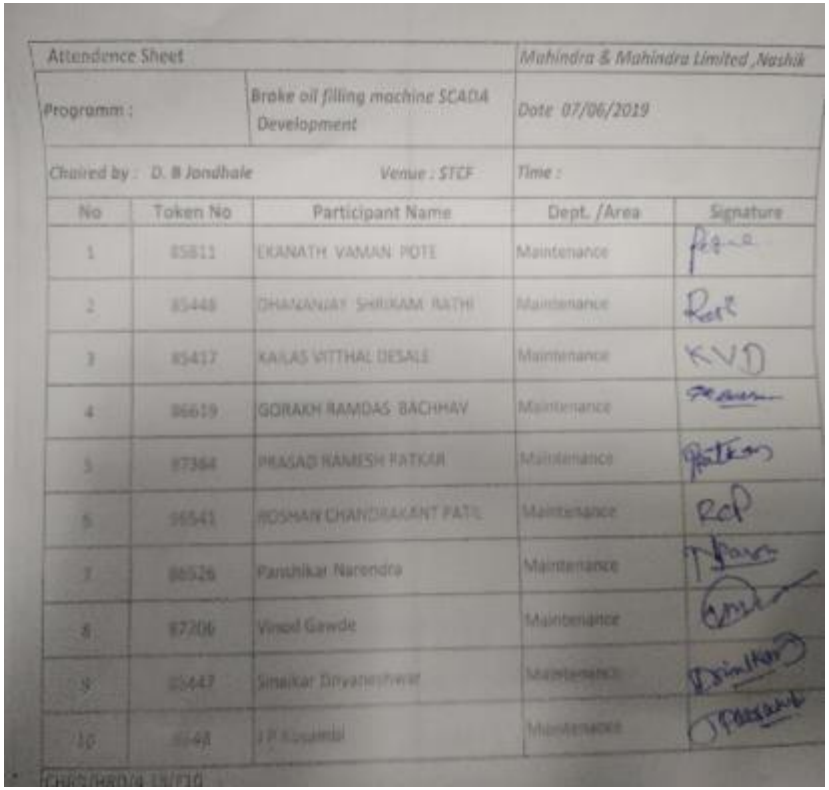
SN	Document	Actual date	Responsibility	Status
4	CLITA Sheet updated	28.08.2019	Dattatray Jondhale	CLOSED



8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training given to Maintenance Associate.	07.08.2019	Dattatray Jondhale	Closed



Attendence Sheet — Mahindra & Mahindra Limited ,Nashik

Programm : Brake oil filling machine SCADA Development — Date 07/06/2019

Chaired by : D. B Jondhale — Venue : STCF — Time :

No	Token No	Participant Name	Dept. /Area	Signature
1	85811	EKANATH VAMAN POTE	Maintenance	
2	85448	DHANANJAY SHRIRAM RATHI	Maintenance	
3	85417	KAILAS VITTHAL DESALE	Maintenance	KVD
4	86619	GORAKH RAMDAS BACHHAV	Maintenance	
5	87384	PRASAD RAMESH PATKAR	Maintenance	
6	96541	ROSHAN CHANDRAKANT PATIL	Maintenance	
7	86526	Panchikar Narendra	Maintenance	
8	87206	Vinod Gawde	Maintenance	
9	85447	Sinaikar Dnyaneshwar	Maintenance	
10	8548	J P Kusumbi	Maintenance	

8. Standardization

8.3 Additional Check -

SN	Document	Actual date	Responsibility	Status
3	Daily Check sheet	28.07.2019	Dattatray Jondhale	CLOSED

Maintenance Inspection Sheet

(Apr'19-Mar'20) Month:Nov.19

Sr. No.	Equipment Name		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
4	BRAKE FILLING MACINE	Condition of Gun seal and leakage from Gun	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O
5		Condition of Diagnostic Connector	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O

9. Summarize & Plan future actions

9.1 Conclusion:

After action implementation no defect reported in last 6 months.

9.2 Horizontal Deployment:

All 12 Filling Machine in STCF.

9.3 Lessons Learnt:

Things Gone Wrong : Initially at the time of Installation failure due to due to Design standardization of machine were not consider.

Things Gone Right :

With SCADA installation done for online cycle base trigger to replace seals has given significant results

Thank you

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